

ALFA RIZI

Junior Python / AI Engineer

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SUMMARY

Entry-level Software Engineer focused on Python backend development and AI-powered applications, with hands-on experience building asynchronous FastAPI services, RAG pipelines, vector search, document processing, rate limiting, and modern web applications.

TECHNICAL SKILLS

Languages: Python, SQL, TypeScript, JavaScript, Java

AI & Backend: FastAPI, ChromaDB, Gemini, PyMuPDF, Express.js

Web: React, Next.js, Vite, Tailwind CSS

Infrastructure & Tools: Docker, PostgreSQL, Redis, Git, GitHub Actions, Linux

PROFESSIONAL EXPERIENCE

Lead Intern — IT & Business Operations | PT Resik Cemerlang (Pasim Group), Bandung | 2025–Present

- Led a 5-person intern team supporting daily IT operations, hardware diagnostics, and workstation SLA compliance.
- Managed OS security patching, software configuration, and antivirus definition updates across corporate laptops.
- Validated technical specifications across thousands of medical device asset records, maintaining database integrity and audit readiness.

LEADERSHIP & ORGANIZATIONS

Academic Staff — HIMAMI

- Coordinated academic development activities, peer-to-peer coding support, and technical learning initiatives.

SELECTED ENGINEERING PROJECTS

StudyMate AI — Asynchronous RAG Document Knowledge Assistant

Python 3.12, FastAPI, ChromaDB, Gemini 2.0 Flash, Docker | 2026

- Built an asynchronous RAG pipeline that parses textbook PDFs with PyMuPDF, performs vector search with ChromaDB, and returns exact page citations.
- Integrated Gemini 2.0 Flash for summaries, quizzes, and 3D flashcards, with a 60 requests/minute token-bucket rate limiter.

GitHub: github.com/a1fariz/StudyMateAI

ReLoop — Enterprise Circular-Commerce Platform & Escrow Architecture

Java 17, Quarkus 3.15 LTS, Next.js 14, PostgreSQL 16, Redis 7 | 2026

- Architected a modular monolith across 25 bounded contexts, enforcing module boundaries with ArchUnit and asynchronous processing through a transactional outbox worker.
- Engineered a double-entry escrow ledger and transaction-safe checkout leases using PostgreSQL pessimistic locking (SELECT ... FOR UPDATE) to prevent concurrent inventory conflicts.
- Integrated Next.js with TanStack Query and Redis catalog caching for responsive product browsing and inventory-aware checkout.

Demo: reloop.biz.id | GitHub: github.com/a1fariz/ReLoop

Vanguard IELTS — Academic Preparation & SRS Platform

React 19, TypeScript, Vite, Tailwind CSS v4, Web Speech API | 2026

- Developed an IELTS preparation suite covering Listening, Reading, Writing, and Speaking across CEFR A1–C1 levels.
- Implemented a SuperMemo-inspired spaced-repetition vocabulary engine and Shadowing Studio using native Web Speech synthesis.

Demo: vanguard-ielts.vercel.app | GitHub: github.com/a1fariz/Vanguard-IELTS

EDUCATION

Universitas Nasional PASIM — Associate Degree in Informatics Management (D3) | Expected May 2027 | GPA: 3.6/4.0

PUB Intensive Technical Scholarship — Logic & C, Data Structures, Relational Databases, Advanced Java Development, and React Development | 2024–Present

CERTIFICATIONS & TECHNICAL TRAINING

- Python for Everybody Specialization — University of Michigan (2026)
- Programming for Data Science with Python — HarvardX (2025)
- Certified Entry-Level Python Programmer (PCEP) — Python Institute (2025)
- Introduction to Cloud Computing — IBM (2026)
- Git and GitHub Training — PUB Programming Training Center, PASIM National University (Jan 2026)
- Database Training — PUB Programming Training Center, PASIM National University (May 2025)
- Data Structure Training — PUB Programming Training Center, PASIM National University (May 2025)
- Web Programming Training (HTML, CSS, JavaScript) — PUB Programming Training Center, PASIM National University (Aug 2025)